Name:

Rates of Chemical Reactions

The speed at which a chemical reaction occurs is called the _____ of _____. _____ are an example of a reaction occurring quickly. _____ is an example of a reaction that occurs slowly.

Factors that affect the rate of a reaction include:

-
-
-
-
-
-

You will conduct the following reaction under different conditions to see how they affect the rate of reaction. Write the balanced formula equation below the word equation.

Magnesium + Sulfuric Acid $\longrightarrow$ Magnesium Sulfate + Hydrogen Gas

Control

A control is an experiment you do to use as a comparison. For this experiment you will not change any of the above factors.

Why is it important to have a control?

_____

_____

- ✓ Put 2cm of 2M H_2SO_4 in a test tube. Place the test tube in a test tube rack.
- ✓ Get your timer ready.
- ✓ Start the timer when you place your magnesium strip in the acid.
- ✓ Stop the timer when the magnesium has all reacted.
- ✓ Record your results in the table.

Temperature

- ✓ Read the temperature section on page 163
- ✓ Half fill a beaker with boiling water from the kettle.
- ✓ Half fill another beaker with iced water
- ✓ Put 2cm of 2M Sulfuric acid in the bottom of two test tubes.

How do you think the reaction rates of the hot and cold test tubes will compare to the room temperature one you did previously?

_____

_____

What is your independent variable for this experiment? _____

_____

What is your dependent variable? _____

Write a hypothesis

_____

_____

- ✓ Place one of these test tubes in the beaker of boiling water and the other in the beaker of iced water.
- ✓ Get your timer ready.
- ✓ Start your timer when you place the strips of magnesium in the acid.
- ✓ Record how long it takes the magnesium to disappear from each test tube on the table.

Write a conclusion regarding the effect of temperature on reaction rate.

_____

_____

_____

_____

_____

Concentration

Read the concentration section on page164

The more concentrated a solution the _____ the rate of reaction.

Use a diagram to explain why this is the case.

The concentration of acids is measured in moles (M). The higher the number the more concentrated the acid is. In the previous experiments the concentration of the acid used was _____M, in this experiment we will use 0.2M sulphuric acid. 0.2M H_2SO_4 is more _____ than 2M H_2SO_4.

Write a hypothesis to show what you expect to happen in this experiment compared with the control.

_____

_____

- ✓ Put 2cm of 0.2M H_2SO_4 in the bottom of a test tube. Place the test tube in a test tube rack.
- ✓ Get your timer ready.
- ✓ Start the timer when you place your magnesium strip in the acid.
- ✓ Stop the timer when the magnesium has all reacted.
- ✓ Record your results in the table.

Did your results support your hypothesis? Use figures from your results.

_____

_____

_____

Agitation

Read the agitation section on page 164.

What does agitation mean?

_____

_____

Place 2cm of 2M Sulfuric acid in the bottom of a test tube.

Do you think this piece of magnesium will dissolve faster if you swirl the test tube? Why or why not?

_____

_____

- ✓ Get your timer ready.
- ✓ Start your timer when you place the strip of magnesium in the acid.
- ✓ Swirl the test tube around taking care not to spill any.
- ✓ Record how long it takes the magnesium to disappear from the test tube.
- ✓ Record your results on the table.

Surface area

Read the surface area section on page 165.

What is surface area?

_____

_____

How would you increase the surface area of a substance?

_____

_____

Does increasing the surface area increase or decrease the rate of reaction? Use a diagram to explain why?

_____

_____

_____

- ✓ Place 2cm of 2M Sulfuric acid in the bottom of a test tube. Place the test tube in a test tube rack.
- ✓ Cut a strip of magnesium in to small pieces. Has this increased or decreased the surface area of the magnesium? _____ _____

Will this piece of magnesium will dissolve faster or slower than the control?

_____

- ✓ Get your timer ready.
- ✓ Start your timer when you place the pieces of magnesium in the acid.
- ✓ Record how long it takes the magnesium to disappear from the test tube.
- ✓ Record your results on the table.

Results Table

Test conditions	Time
Magnesium at room temperature	
Magnesium in hot environment	
Magnesium in very cold environment	
Magnesium in dilute acid (0.2M)	
Agitated while reacting	
Magnesium cut into pieces	

Look at your results table. Carefully compare each of your results with the control.

Create a summary to show how each of the factors you tested affects reaction rate:

There are two other factors that can affect reaction rate that we did not test today.

Catalysts

Read the catalyst section on page 165.

What is a catalyst?

_____

_____

Explain the two ways in which catalysts work:

1. _____

   _____

2. _____

   _____

Enzymes

Read the enzyme section on page 165.

What are enzymes?

_____

_____

Organisms use enzymes to speed up chemical reactions within their body.

Amylase is an enzyme found in _____. It is used to _____

_____

_____